

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

5th Semester of B. Pharm. Examination

University Theory Examination April 2016

PH306 Pharmaceutical Biotechnology

Date: 19.04.16 , Tuesday

Time: 10:00 a.m to 1:00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

- Q 1 Attempt any TWO of the following**
- | | | |
|---|---|----|
| A | Write a note on protoplast fusion. What are the applications of it? | 10 |
| B | Discuss the double helical DNA structure and Replication of DNA | 10 |
| C | i) Explain different types of mutations.
ii) Explain in detail mutagens. | 10 |
- Q 2 Attempt any TWO of the following**
- | | | |
|---|--|----------|
| A | a) What is Gene cloning? Explain gene cloning for larger fragment of DNA.
b) Explain PCR reaction in detail | 05
05 |
| B | Discuss methods for isolation of mutants. | 10 |
| C | Write a note on
i) Transformation and ii) Conjugation | 10 |

SECTION- II

- Q 3 Attempt any TWO of the following**
- | | | |
|---|--|----------|
| A | Write a note on following
a) Adsorption method for enzyme immobilization
b) entrapment method | 05
05 |
| B | Enlist the methods for production of Vinegar. Explain submerged fermentation method with a neat diagram of fermentor | 10 |
| C | Write the pathway of itaconic acid and Kojic acid production by biotransformation way | 10 |
- Q 4 Attempt any TWO of the following**
- | | | |
|---|--|----|
| A | How do a technical person take precaution while working in biotechnology laboratory? | 10 |
| B | What is single cell protein? Write its application and salient features for production of it | 10 |
| C | Explain the pathway and production of penicillin by microbial biotransformation way | 10 |